

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Sulfur dioxide (so2)
Observed / expected baseline	7.2 / 0.217241 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-08-03 15:00 UTC
Location	29.6697, -95.1285
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	7 / 706	20.95 to 100; mean 68.7606	70.53 at 2026-08-03 14:55Z; 5 min before event
asos	Air temperature (temperature)	degC	7 / 706	23.8889 to 37; mean 30.228	31 at 2026-08-03 14:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	7 / 882	0 to 360; mean 154.603	260 at 2026-08-03 14:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	7 / 919	0 to 33.4389; mean 2.8728	3.08666 at 2026-08-03 14:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	8 / 96	5875.21 to 5918.17; mean 5896.28	5880.42 at 2026-08-03 12:00Z; 3 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	8 / 96	11.2803 to 2539.72; mean 842.081	164.72 at 2026-08-03 12:00Z; 3 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	8 / 96	27.1461 to 58.478; mean 37.8292	32.8749 at 2026-08-03 12:00Z; 3 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	8 / 96	1005.04 to 1013.6; mean 1009.42	1009.77 at 2026-08-03 12:00Z; 3 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	8 / 96	18.2063 to 21.4418; mean 19.5371	18.5016 at 2026-08-03 12:00Z; 3 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	8 / 96	-3.62395 to 3.87893; mean 0.0183	0.82968 at 2026-08-03 12:00Z; 3 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	8 / 96	-3.17918 to 5.58564; mean 1.4632	1.02434 at 2026-08-03 12:00Z; 3 h before event
openaq	Ozone (ozone)	ppm	17 / 1119	-0.004 to 0.081; mean 0.0281	0.017 at 2026-08-03 15:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 214	12 to 172; mean 39.2757	40 at 2026-08-03 15:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	72 / 2856	0.8 to 50.8; mean 11.0942	14.1 at 2026-08-03 15:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	0 to 100; mean 26.5639	0 at 2026-08-03 15:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 360	34 to 95; mean 70.0972	76 at 2026-08-03 15:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 0.25; mean 0.0028	0 at 2026-08-03 15:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 360	1007 to 1014; mean 1011.17	1013 at 2026-08-03 15:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 360	25.09 to 37.65; mean 31.0794	31.13 at 2026-08-03 15:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 355; mean 170.589	211 at 2026-08-03 15:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.26 to 6.47; mean 2.1436	1.26 at 2026-08-03 15:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	66 / 4555	0 to 557.7; mean 13.1328	12.4 at 2026-08-03 15:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-08-03 18:19Z; 3 h 20 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-03 18:50Z; 3 h 50 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	7 / 7	0.0277646 to 0.0328455; mean 0.031	0.0304726 at 2026-08-03 18:19Z; 3 h 20 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-08-03 18:19Z; 3 h 20 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000121637 to 0.000318768; mean 0.0003	0.00030109 at 2026-08-03 18:50Z; 3 h 50 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-03 18:50Z; 3 h 50 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	3.75274e-05 to 5.22023e-05; mean 0	4.72661e-05 at 2026-08-03 18:50Z; 3 h 50 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-08-03 18:50Z; 3 h 50 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-03 18:50Z; 3 h 50 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-6.07222e-05 to 0.000178122; mean 0	3.30053e-05 at 2026-08-03 18:50Z; 3 h 50 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-03 18:50Z; 3 h 50 min after event
tceq	Carbon monoxide (co)	ppm	3 / 178	0 to 0.9; mean 0.227	0.2 at 2026-08-03 15:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 717	-2.5 to 33.6; mean 4.6685	7.2 at 2026-08-03 15:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 177	-0.5 to 7.2; mean -0.0271	7.2 at 2026-08-03 15:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	08-02 00Z 82.19, 06Z 87.03, 12Z 61.5, 18Z 36.57, 08-03 00Z 69.27, 06Z 88.24, 12Z 68.41, 18Z 46.44, 08-04 00Z 75.34, 06Z 90.78, 12Z 70.42, 18Z 53.5, 08-05 00Z 75
asos	Air temperature (temperature)	08-02 00Z 28.91, 06Z 27.92, 12Z 31.06, 18Z 35.13, 08-03 00Z 29.55, 06Z 26.34, 12Z 30.64, 18Z 34.39, 08-04 00Z 28.59, 06Z 25.56, 12Z 30.45, 18Z 33.83, 08-05 00Z 29.45
asos	Wind direction (wind_direction)	08-02 00Z 225.9, 06Z 235.6, 12Z 118.6, 18Z 158.1, 08-03 00Z 162.3, 06Z 135.1, 12Z 162.6, 18Z 151.9, 08-04 00Z 157.8, 06Z 70.82, 12Z 136.5, 18Z 163.2, 08-05 00Z 182.6
asos	Wind speed (wind_speed)	08-02 00Z 3.384, 06Z 2.798, 12Z 2.045, 18Z 2.86, 08-03 00Z 2.839, 06Z 1.488, 12Z 2.275, 18Z 4.623, 08-04 00Z 3.374, 06Z 0.8376, 12Z 2.018, 18Z 5.535, 08-05 00Z 4.313
noaa_gfs	500-hPa geopotential height (gh_500)	08-02 06Z 5901, 12Z 5881, 18Z 5886, 08-03 00Z 5877, 06Z 5887, 12Z 5880, 18Z 5896, 08-04 00Z 5900, 06Z 5914, 12Z 5900, 18Z 5916, 08-05 00Z 5918
noaa_gfs	Planetary boundary layer height (pbl_height)	08-02 06Z 330, 12Z 205.3, 18Z 2122, 08-03 00Z 932.4, 06Z 167.6, 12Z 95.48, 18Z 2217, 08-04 00Z 938.1, 06Z 301.5, 12Z 101.9, 18Z 2014, 08-05 00Z 679.3
noaa_gfs	Precipitable water (precipitable_water)	08-02 06Z 53.23, 12Z 41.39, 18Z 32.8, 08-03 00Z 30.37, 06Z 31.42, 12Z 33.08, 18Z 34.9, 08-04 00Z 38.49, 06Z 33.44, 12Z 35.65, 18Z 39.53, 08-05 00Z 49.65
noaa_gfs	Surface pressure (surface_pressure)	08-02 06Z 1008, 12Z 1009, 18Z 1009, 08-03 00Z 1006, 06Z 1009, 12Z 1010, 18Z 1010, 08-04 00Z 1008, 06Z 1010, 12Z 1011, 18Z 1012, 08-05 00Z 1011
noaa_gfs	850-hPa temperature (t_850)	08-02 06Z 20.55, 12Z 20.49, 18Z 19.32, 08-03 00Z 20.6, 06Z 19.23, 12Z 18.47, 18Z 18.84, 08-04 00Z 19.88, 06Z 20, 12Z 19.02, 18Z 18.84, 08-05 00Z 19.2
noaa_gfs	10-m eastward wind (u_10m)	08-02 06Z 3.07, 12Z 1.21, 18Z -0.8204, 08-03 00Z -1.848, 06Z 1.145, 12Z 1.115, 18Z -1.169, 08-04 00Z -2.435, 06Z 0.7441, 12Z 0.6063, 18Z -1.129, 08-05 00Z -0.2686
noaa_gfs	10-m northward wind (v_10m)	08-02 06Z 1.743, 12Z -1.597, 18Z -2.622, 08-03 00Z -0.4377, 06Z 2.214, 12Z 0.8743, 18Z 1.766, 08-04 00Z 4.523, 06Z 2.888, 12Z 1.277, 18Z 2.549, 08-05 00Z 4.38
openaq	Ozone (ozone)	08-02 00Z 0.01671, 06Z 0.008851, 12Z 0.024, 18Z 0.05939, 08-03 00Z 0.05227, 06Z 0.02463, 12Z 0.02048, 18Z 0.05143, 08-04 00Z 0.02236, 06Z 0.009273, 12Z 0.01327, 18Z 0.02896, 08-05 00Z 0.0184
openaq	PM10 (pm10)	08-02 00Z 25.22, 06Z 22.22, 12Z 22.89, 18Z 16.61, 08-03 00Z 20.44, 06Z 20.5, 12Z 50.18, 18Z 43.35, 08-04 00Z 41.67, 06Z 40.67, 12Z 73.56, 18Z 76.56, 08-05 00Z 61.78
openaq	PM2.5 (pm25)	08-02 00Z 10.37, 06Z 10, 12Z 8.052, 18Z 8.638, 08-03 00Z 10.86, 06Z 9.387, 12Z 9.881, 18Z 12.12, 08-04 00Z 11.68, 06Z 15.27, 12Z 14.76, 18Z 11.79, 08-05 00Z 10.37
openweather	Cloud cover (cloud_cover)	08-02 00Z 47.19, 06Z 67.31, 12Z 85.45, 18Z 32.53, 08-03 00Z 19.73, 06Z 1.069, 12Z 0, 18Z 2.1, 08-04 00Z 23.13, 06Z 1.8, 12Z 0.3333, 18Z 23, 08-05 00Z 74.67
openweather	Relative humidity (humidity)	08-02 00Z 76.88, 06Z 85.97, 12Z 70.94, 18Z 41.77, 08-03 00Z 65.2, 06Z 83.55, 12Z 75.97, 18Z 51.1, 08-04 00Z 71.7, 06Z 87.3, 12Z 77.63, 18Z 56.21, 08-05 00Z 70.93
openweather	Precipitation rate (precipitation)	08-02 00Z 0, 06Z 0.007586, 12Z 0, 18Z 0, 08-03 00Z 0, 06Z 0, 12Z 0, 18Z 0, 08-04 00Z 0, 06Z 0, 12Z 0.01, 18Z 0, 08-05 00Z 0.03267
openweather	Surface pressure (pressure)	08-02 00Z 1010, 06Z 1010, 12Z 1012, 18Z 1009, 08-03 00Z 1009, 06Z 1011, 12Z 1012, 18Z 1010, 08-04 00Z 1011, 06Z 1012, 12Z 1014, 18Z 1013, 08-05 00Z 1013
openweather	Air temperature (temperature)	08-02 00Z 30.23, 06Z 28.4, 12Z 30.66, 18Z 36.36, 08-03 00Z 31.37, 06Z 27.22, 12Z 30.23, 18Z 35.93, 08-04 00Z 30.17, 06Z 26.51, 12Z 29.98, 18Z 35.38, 08-05 00Z 31.24
openweather	Wind direction (wind_direction)	08-02 00Z 225.6, 06Z 194.6, 12Z 90.61, 18Z 166.6, 08-03 00Z 158.4, 06Z 175.3, 12Z 215.5, 18Z 151.8, 08-04 00Z 161, 06Z 175, 12Z 192.8, 18Z 163.8, 08-05 00Z 177.9
openweather	Wind speed (wind_speed)	08-02 00Z 2.742, 06Z 2.043, 12Z 1.942, 18Z 2.036, 08-03 00Z 1.967, 06Z 1.916, 12Z 1.322, 18Z 2.397, 08-04 00Z 2.572, 06Z 1.775, 12Z 1.748, 18Z 3.054, 08-05 00Z 3.223

Source	Metric	Values
purpleair	PM2.5 (pm25)	08-02 00Z 14.91, 06Z 12.09, 12Z 10.19, 18Z 9.365, 08-03 00Z 12.34, 06Z 11.94, 12Z 12.27, 18Z 13.73, 08-04 00Z 13.27, 06Z 17.57, 12Z 15.78, 18Z 13.68, 08-05 00Z 15.47
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	08-02 18Z 1, 08-03 18Z 1, 08-04 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	08-02 18Z 1, 08-03 18Z 1, 08-04 18Z 1
sentinel5p	CO column (s5p_co_column)	08-02 18Z 0.03284, 08-03 18Z 0.03095, 08-04 18Z 0.02776
sentinel5p	CO granules available (s5p_co_granule_available)	08-02 18Z 1, 08-03 18Z 1, 08-04 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	08-02 18Z 0.0003127, 08-03 18Z 0.0002755, 08-04 18Z 0.0001216
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	08-02 18Z 1, 08-03 18Z 1, 08-04 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	08-02 18Z 4.15e-05, 08-03 18Z 4.973e-05, 08-04 18Z 3.753e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	08-02 18Z 1, 08-03 18Z 1, 08-04 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	08-02 18Z 1, 08-03 18Z 1, 08-04 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	08-02 18Z -4.61e-05, 08-03 18Z 0.0001056, 08-04 18Z -3.371e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	08-02 18Z 1, 08-03 18Z 1, 08-04 18Z 1
tceq	Carbon monoxide (co)	08-02 00Z 0.5, 06Z 0.1167, 12Z 0.1333, 18Z 0.16, 08-03 00Z 0.2444, 06Z 0.2389, 12Z 0.2722, 18Z 0.25, 08-04 00Z 0.2111, 06Z 0.2556, 12Z 0.3167, 18Z 0.2615, 08-05 00Z 0.3
tceq	Nitrogen dioxide (no2)	08-02 00Z 4, 06Z 3.036, 12Z 3.271, 18Z 2.811, 08-03 00Z 4.778, 06Z 8.049, 12Z 7.837, 18Z 4.965, 08-04 00Z 2.549, 06Z 4.78, 12Z 6.113, 18Z 3.278, 08-05 00Z 2.103
tceq	Sulfur dioxide (so2)	08-02 06Z -0.07778, 12Z -0.09444, 18Z -0.02778, 08-03 00Z -0.03333, 06Z -0.2056, 12Z 0.7444, 18Z -0.07222, 08-04 00Z -0.04444, 06Z -0.2556, 12Z -0.16, 18Z -0.1077, 08-05 00Z -0.1

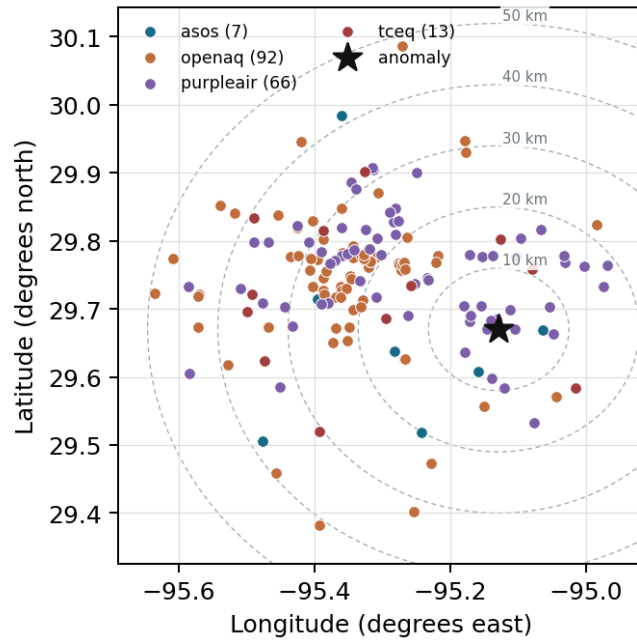
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

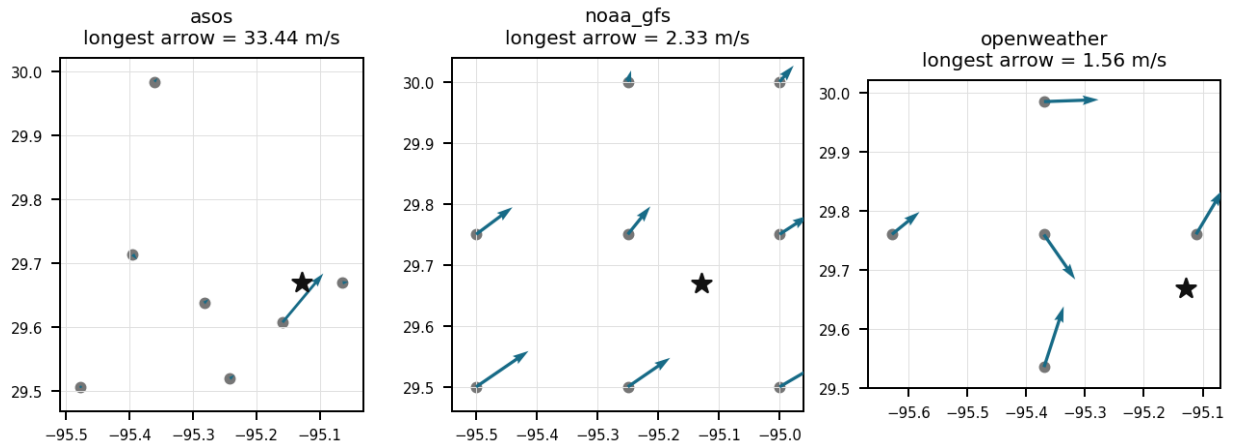
Source	Metric	Values
asos	Relative humidity (humidity)	T41 (6.225 km) 71.26, EFD (7.53 km) 68.88, HOU (15.295 km) 69.06, LVJ (20.026 km) 71.43, MCJ (26.207 km) 66.4, AXH (38.286 km) 68.88, IAH (41.543 km) 65.22
asos	Air temperature (temperature)	T41 (6.225 km) 29.95, EFD (7.53 km) 30.43, HOU (15.295 km) 30.51, LVJ (20.026 km) 29.98, MCJ (26.207 km) 30.54, AXH (38.286 km) 29.84, IAH (41.543 km) 30.73
asos	Wind direction (wind_direction)	T41 (6.225 km) 172.7, EFD (7.53 km) 146.6, HOU (15.295 km) 182.8, LVJ (20.026 km) 141.4, MCJ (26.207 km) 178.1, AXH (38.286 km) 94.01, IAH (41.543 km) 167.4
asos	Wind speed (wind_speed)	T41 (6.225 km) 3.367, EFD (7.53 km) 3.819, HOU (15.295 km) 3.472, LVJ (20.026 km) 2.219, MCJ (26.207 km) 3.824, AXH (38.286 km) 1.058, IAH (41.543 km) 2.894
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (14.741 km) 5896, gfs:29.75,-95.00 (15.288 km) 5896, gfs:29.50,-95.25 (22.231 km) 5896, gfs:29.50,-95.00 (22.598 km) 5896, gfs:29.75,-95.50 (36.97 km) 5897, gfs:30.00,-95.25 (38.548 km) 5896, gfs:30.00,-95.00 (38.76 km) 5896, gfs:29.50,-95.50 (40.577 km) 5897
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (14.741 km) 1015, gfs:29.75,-95.00 (15.288 km) 843.6, gfs:29.50,-95.25 (22.231 km) 751.6, gfs:29.50,-95.00 (22.598 km) 732.4, gfs:29.75,-95.50 (36.97 km) 1115, gfs:30.00,-95.25 (38.548 km) 919.3, gfs:30.00,-95.00 (38.76 km) 574.1, gfs:29.50,-95.50 (40.577 km) 785.5

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (14.741 km) 37.99, gfs:29.75,-95.00 (15.288 km) 37.5, gfs:29.50,-95.25 (22.231 km) 39.7, gfs:29.50,-95.00 (22.598 km) 38.81, gfs:29.75,-95.50 (36.97 km) 37.23, gfs:30.00,-95.25 (38.548 km) 36.6, gfs:30.00,-95.00 (38.76 km) 35.7, gfs:29.50,-95.50 (40.577 km) 39.11
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (14.741 km) 1010, gfs:29.75,-95.00 (15.288 km) 1010, gfs:29.50,-95.25 (22.231 km) 1010, gfs:29.50,-95.00 (22.598 km) 1011, gfs:29.75,-95.50 (36.97 km) 1008, gfs:30.00,-95.25 (38.548 km) 1008, gfs:30.00,-95.00 (38.76 km) 1009, gfs:29.50,-95.50 (40.577 km) 1009
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (14.741 km) 19.52, gfs:29.75,-95.00 (15.288 km) 19.4, gfs:29.50,-95.25 (22.231 km) 19.54, gfs:29.50,-95.00 (22.598 km) 19.46, gfs:29.75,-95.50 (36.97 km) 19.79, gfs:30.00,-95.25 (38.548 km) 19.54, gfs:30.00,-95.00 (38.76 km) 19.33, gfs:29.50,-95.50 (40.577 km) 19.71
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (14.741 km) -0.01159, gfs:29.75,-95.00 (15.288 km) -0.06992, gfs:29.50,-95.25 (22.231 km) 0.04008, gfs:29.50,-95.00 (22.598 km) 0.09258, gfs:29.75,-95.50 (36.97 km) 0.4201, gfs:30.00,-95.25 (38.548 km) -0.5749, gfs:30.00,-95.00 (38.76 km) 0.1242, gfs:29.50,-95.50 (40.577 km) 0.1259
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (14.741 km) 1.355, gfs:29.75,-95.00 (15.288 km) 1.25, gfs:29.50,-95.25 (22.231 km) 1.975, gfs:29.50,-95.00 (22.598 km) 2.321, gfs:29.75,-95.50 (36.97 km) 1.516, gfs:30.00,-95.25 (38.548 km) 0.7906, gfs:30.00,-95.00 (38.76 km) 0.9381, gfs:29.50,-95.50 (40.577 km) 1.561
openaq	Ozone (ozone)	332 (0.032 km) 0.03064, 2800 (11.598 km) 0.02581, 2039 (14.096 km) 0.02672, 320 (14.272 km) 0.02529, 3378 (14.356 km) 0.02747, 5077791 (14.558 km) 0.02245, 3214 (14.765 km) 0.02819, 325 (16.163 km) 0.0071, +9 more
openaq	PM10 (pm10)	271 (0.032 km) 35.4, 13380082 (14.356 km) 55.83, 15852419 (21.358 km) 26.49
openaq	PM2.5 (pm25)	268 (0.032 km) 14.22, 13955815 (12.727 km) 9.399, 14404954 (13.725 km) 11.22, 14157975 (14.275 km) 13.94, 3379 (14.356 km) 12.55, 15539682 (14.359 km) 11.43, 8967123 (14.413 km) 12.75, 8967479 (14.413 km) 11.25, +64 more
openweather	Cloud cover (cloud_cover)	grid:east (10.221 km) 25.76, grid:center (25.388 km) 26.74, grid:south (27.676 km) 33.1, grid:north (42.079 km) 20.67, grid:west (49.325 km) 26.56
openweather	Relative humidity (humidity)	grid:east (10.221 km) 74.18, grid:center (25.388 km) 68.4, grid:south (27.676 km) 71.43, grid:north (42.079 km) 67.14, grid:west (49.325 km) 69.33
openweather	Precipitation rate (precipitation)	grid:east (10.221 km) 0.005, grid:center (25.388 km) 0.003472, grid:south (27.676 km) 0, grid:north (42.079 km) 0.005556, grid:west (49.325 km) 0
openweather	Surface pressure (pressure)	grid:east (10.221 km) 1011, grid:center (25.388 km) 1011, grid:south (27.676 km) 1011, grid:north (42.079 km) 1011, grid:west (49.325 km) 1011
openweather	Air temperature (temperature)	grid:east (10.221 km) 30.73, grid:center (25.388 km) 31.61, grid:south (27.676 km) 30.77, grid:north (42.079 km) 31.18, grid:west (49.325 km) 31.12
openweather	Wind direction (wind_direction)	grid:east (10.221 km) 156.2, grid:center (25.388 km) 192, grid:south (27.676 km) 194.4, grid:north (42.079 km) 131.2, grid:west (49.325 km) 179.1
openweather	Wind speed (wind_speed)	grid:east (10.221 km) 3.233, grid:center (25.388 km) 2.042, grid:south (27.676 km) 2.019, grid:north (42.079 km) 1.983, grid:west (49.325 km) 1.441
purpleair	PM2.5 (pm25)	128007 (1.625 km) 6.328, 161689 (1.727 km) 13.01, 222589 (1.859 km) 13.72, 268165 (2.296 km) 14.82, 222593 (3.655 km) 13.1, 186263 (4.264 km) 15.18, 98633 (4.643 km) 6.389, 244939 (4.684 km) 12.15, +58 more
tceq	Carbon monoxide (co)	482011039 (0.0 km) 0.1787, 482011035 (14.355 km) 0.1722, 482011052 (29.756 km) 0.3206
tceq	Nitrogen dioxide (no2)	482011039 (0.0 km) 2.962, 482011015 (10.998 km) 6.156, 482011035 (14.355 km) 7.971, 482011050 (14.566 km) 1.3, 482010026 (14.787 km) 6.685, 482010416 (16.162 km) 4.638, 482011052 (29.756 km) 12.24, 480391004 (30.447 km) 2.206, +4 more
tceq	Sulfur dioxide (so2)	482011039 (0.0 km) 0.379, 482011035 (14.355 km) -0.3423, 482010051 (33.805 km) -0.1667

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

A short-duration plume from nearby industrial sulfur sources is the most likely physical cause of the 7.2 ppb sulfur dioxide spike at the Houston monitor because sulfur dioxide rose sharply at one TCEQ site while nearby sulfur dioxide monitors did not show a comparable regional increase, which is consistent with a localized source impact rather than a citywide event.

Mark exactly one:

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Note (optional):

Claim 2 of 36

Combustion-related industrial operations such as refinery process upset, sulfur recovery activity, or stack emissions are more plausible than wildfire or broad urban traffic as the sulfur dioxide source type because carbon monoxide, ozone, and particulate matter were not simultaneously extreme near the monitor while satellite sulfur dioxide later that day showed only a modest column enhancement compatible with a small or transient local plume.

Mark exactly one:

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Note (optional):

Claim 3 of 36

A severe thunderstorm or tornado event in the Houston area has stirred up dust and debris, contributing to the air quality anomaly.

Mark exactly one:

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Note (optional):

Claim 4 of 36

At the exact time and location of the sulfur dioxide anomaly on 2026-08-03T15:00:00+00:00, TCEQ measured concurrent nitrogen dioxide levels of 7.2 ppb, while OpenAQ reported PM10 levels of 40.0 ug/m3 and PM2.5 levels of 14.1 ug/m3.

Mark exactly one:

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Note (optional):

Claim 5 of 36

Sentinel-5P satellite observations detected an increased mean SO2 column density of 0.0001056 mol/m² over the Houston area on August 3, 2026.

Mark exactly one:

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Note (optional):

Claim 6 of 36

A short-lived local sulfur emission plume affecting the TCEQ monitor is strongly supported because SO₂ reached 7.2 ppb at 2026-08-03T15:00Z at the anomaly monitor while the same monitor's expected value was about 0.22 ppb, the 6-hour TCEQ SO₂ mean for 08-03 12Z rose to 0.744 ppb, and the other two nearby TCEQ SO₂ monitors had much lower window means of -0.342 ppb and -0.167 ppb, which is consistent with a localized rather than region-wide event.

Mark exactly one:

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Note (optional):

Claim 7 of 36

The anomaly is due to a combination of factors including high temperatures, humidity, and stagnant atmospheric conditions, which have led to the formation of ground-level ozone and other pollutants.

Mark exactly one:

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Note (optional):

Claim 8 of 36

The PM_{2.5} levels are elevated, with a mean of 13.1328 ug/m³ and a maximum value of 557.7 ug/m³.

Mark exactly one:

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Note (optional):

Claim 9 of 36

TCEQ ground-level monitoring detected a sharp SO₂ concentration spike of 7.2 ppb at station 482011039 at 15:00 UTC on August 3, 2026, while surrounding TCEQ stations recorded near-zero baseline levels, indicating a highly localized point-source emission event.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 10 of 36**

Meteorological trapping of a nearby plume is supported because the Houston area had light surface winds near the anomaly time of about 1.26 to 3.09 m/s, morning humidity near 70 to 76 percent, clear skies, and a very shallow modeled planetary boundary layer of roughly 95 to 165 m around 12Z, all of which would limit dilution and allow a local SO₂ release to produce a sharp concentration spike.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 11 of 36**

The TCEQ ground monitor 482011039 recorded a peak SO₂ concentration of 7.2 ppb at 15:00 UTC on August 3, 2026.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 12 of 36

A shallow planetary boundary layer height below 200 meters and light surface winds recorded by GFS and ASOS limited atmospheric mixing and enhanced ground-level SO₂ concentrations.

Mark exactly one:

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Note (optional):

Claim 13 of 36

The TCEQ monitor at the anomaly location measured a severe and isolated SO₂ spike of 7.2 ppb at 2026-08-03T15:00:00+00:00 against an expected value near 0.22 ppb.

Mark exactly one:

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Note (optional):

Claim 14 of 36

Sentinel-5P satellite measurements captured elevated total column SO₂ levels over the region on August 3, 2026, providing spaceborne evidence of an atmospheric SO₂ plume.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 1.3181970162090593 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.44980526654126063 m/s; n=96 wind readings); openweather: event wind 1.26 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.45239491837320456 m/s; n=360 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

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Note (optional):

Claim 15 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:

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Note (optional):

Claim 16 of 36

During the 18Z time block on 2026-08-03, Sentinel-5P observed an elevated mean SO2 column density of 0.0001056 mol/m² over the region, compared to negative average column values on August 2 and August 4, 2026.

Mark exactly one:

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Note (optional):

Claim 17 of 36

Localized industrial point-source emissions from regional petrochemical and refining facilities were the primary source of the elevated SO2 concentration detected by TCEQ monitoring instruments.

Mark exactly one:

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Note (optional):

Claim 18 of 36

A short-lived nearby industrial sulfur emission plume is more likely than a broad regional combustion or photochemical event because SO₂ spiked strongly while CO, NO₂, ozone, and PM did not show a comparably extreme co-pollutant enhancement near the monitor.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 19 of 36**

A severe sulfur dioxide spike of 7.2 ppb was recorded by a TCEQ monitor located at coordinates (29.670, -95.129) on 2026-08-03T15:00:00+00:00, representing a z-score of 19.28 relative to the expected baseline of 0.22 ppb.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 20 of 36**

The anomalous pollutant was sulfur dioxide measured by tceq at 29.670, -95.129 in Houston, Texas at 2026-08-03T15:00:00+00:00.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 21 of 36

Houston-area meteorology during the anomaly favored poor dispersion of a nearby plume because local winds were light and the morning boundary layer was shallow.

Mark exactly one:

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Note (optional):

Claim 22 of 36

NOAA GFS model simulations showed a suppressed planetary boundary layer height below 100 meters prior to 12:00 UTC on August 3, while ASOS surface stations recorded low wind speeds averaging 2.28 m/s, creating stagnant early morning conditions that trapped point-source SO₂ near the surface.

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Note (optional):

Claim 23 of 36

Meteorological conditions likely enhanced the sulfur dioxide peak by limiting dilution and allowing narrow plume impacts because low boundary-layer height and light surface winds were present near the event time, with GFS boundary-layer height near 165 m at 12Z and local winds around 1.3 to 3.1 m/s around 15Z.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 1.3181970162090593 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.44980526654126063 m/s; n=96 wind readings); openweather: event wind 1.26 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.45239491837320456 m/s; n=360 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

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Note (optional):

Claim 24 of 36

The CO, SO₂, and O₃ levels were also elevated during the anomaly period, indicating potential contributions from industrial sources and vehicle emissions.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 25 of 36**

The air quality anomaly in Houston, Texas is caused by high levels of SO₂ particles.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 26 of 36**

The air quality anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter and volatile organic compounds.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 27 of 36

The tceq sulfur dioxide measurement at 2026-08-03T15:00:00+00:00 was 7.2 ppb, compared with an expected value of 0.21724137931034482 ppb, corresponding to a z-score of 19.276449828645912 and severe anomaly classification.

Mark exactly one:

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Note (optional):

Claim 28 of 36

A broad combustion or regional photochemical episode is not strongly supported because co-located TCEQ CO was only 0.2 ppm, co-located TCEQ NO2 was 7.2 ppb without an exceptional multi-site enhancement, nearby OpenAQ ozone was only 0.017 ppm at the anomaly time, and Sentinel-5P later showed only modest SO2 column enhancement with no corresponding strong CO enhancement, which argues against a large regional smoke or widespread industrial combustion event even though a larger industrial source cannot be excluded.

Mark exactly one:

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Note (optional):

Claim 29 of 36

The temperature and wind speed data suggest that the air quality anomaly was exacerbated by a heatwave and stagnant atmospheric conditions.

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Note (optional):

Claim 30 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:

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Note (optional):

Claim 31 of 36

Sentinel-5P satellite observations recorded an elevated mean SO₂ column density of 0.000106 mol/m² over the Houston area during the August 3 18:00 UTC granule pass, confirming enhanced atmospheric SO₂ mass relative to negative baseline averages on August 2 and August 4.

Mark exactly one:

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Note (optional):

Claim 32 of 36

The anomaly occurred on August 2nd and 3rd, with the highest levels recorded at 18Z on August 3rd.

Mark exactly one:

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Note (optional):

Claim 33 of 36

The nearest tceq sulfur dioxide monitor had a mean sulfur dioxide value of 0.379 ppb over the context window, and the 6-hour mean for tceq sulfur dioxide at 2026-08-03 12Z was 0.7444 ppb, both far below the 7.2 ppb anomaly value.

Mark exactly one:

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Note (optional):

Claim 34 of 36

The air quality anomaly in Houston, Texas is likely caused by a combination of factors including high levels of PM2.5 and NO2 emissions from industrial activities and vehicle traffic.

Mark exactly one:

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Note (optional):

Claim 35 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:

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Note (optional):

Claim 36 of 36

A shallow morning boundary layer identified by NOAA GFS combined with light surface winds recorded by ASOS stations hindered vertical atmospheric mixing and trapped SO₂ near the surface.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.